arXiv:1901.08846v3 [cs.LG] 29 May 2019

Improving Adversarial Robustness via Promoting Ensemble Diversity

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Abstract

Though deep neural networks have achieved significant progress on various tasks, often enhanced by model ensemble, existing high-performance models can be vulnerable to adversarial attacks. Many efforts have been devoted to enhancing the robustness of individual networks and then constructing a straightforward ensemble, e.g., by directly averaging the outputs, which ignores the interaction among networks. This paper presents a new method that explores the interaction among individual networks to improve robustness for ensemble models. Technically, we define a new notion of ensemble diversity in the adversarial setting as the diversity among *non-maximal* predictions of individual members, and present an adaptive diversity promoting (ADP) regularizer to encourage the diversity, which leads to globally better robustness for the ensemble by making adversarial examples difficult to transfer among individual members. Our method is computationally efficient and compatible with the defense methods acting on individual networks. Empirical results on various datasets verify that our method can improve adversarial robustness while maintaining state-of-the-art accuracy on normal examples.

1. Introduction

Deep neural networks (DNNs) have recently achieved significant progress in many tasks, such as image classification (Russakovsky et al., 2015) and speech recognition (Graves & Jaitly, 2014). However, they are not problemless. Recent work has shown that high-performance classifiers are vulnerable to adversarial attacks (Goodfellow et al., 2015; Papernot et al., 2016b; Carlini & Wagner, 2017b; Athalye et al., 2018), where images with human imperceptible perturbations (i.e., adversarial examples) can be generated to fool the classifiers. To improve the adversarial robustness of classifiers, various kinds of defenses have been proposed (Kurakin et al., 2017b; 2018a; Kannan et al., 2018), and one practically effective strategy is to construct ensembles of the enhanced networks to obtain stronger defenses (Liao et al., 2018; Kurakin et al., 2018b).

However, most of these defense strategies focus on enhancing an individual network, while ignoring the potentially very informative interactions among multiple models. As widely observed, adversarial examples can have strong transferability among models (Papernot et al., 2016a; Kurakin et al., 2018b). When training an ensemble model, the ignorance of interaction among individual members may lead them to return similar predictions or feature representations (Dauphin et al., 2014; Li et al., 2016). Since it is easier for adversarial examples to transfer among similar models, it is probable for the adversarial examples crafted for one individual member to also fool other members, or even fool the whole ensemble model (See experiments).

Previous work has shown that for a single network, promoting the diversity among learned features of different classes can improve adversarial robustness (Pang et al., 2018a;b). In this paper, we propose to improve adversarial robustness from a new perspective by promoting the diversity among the predictions returned by different members of an ensemble, called *ensemble diversity*. Our approach is orthogonal to other defenses acting on a single network and thus is compatible with most of the previous defense methods.

Technically, we first develop a new notion of ensemble diversity in the adversarial setting, which is significantly different from the previous attempts in the standard setting of ensembling weak classifiers in a non-adversarial environment. Namely, previous work has defined ensemble diversity by considering the prediction errors of individual classifiers in the ensemble (Liu & Yao, 1999a;b; Dietterich, 2000; Islam et al., 2003). However, this definition is inappropriate for our case since most DNNs can achieve high accuracy on a training set (Goodfellow et al., 2016) and promoting diversity of prediction errors will largely sacrifice accuracy. Instead, we define the diversity on the *non-maximal* predictions of each network (detailed in Eq. (3)). Geometrically, the diversity

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Proceedings of the 36th International Conference on Machine Learning, Long Beach, California, PMLR 97, 2019. Copyright 2019 by the author(s).

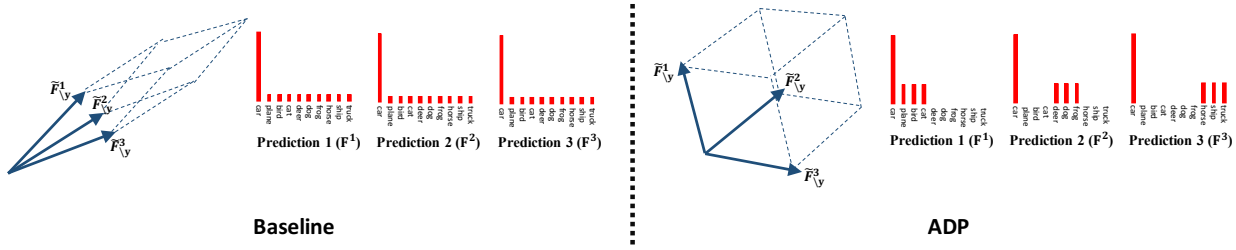

Figure 1. Illustration of the ensemble diversity. **Baseline:** Individually training each member of the ensemble. **ADP:** Simultaneously training all the members of the ensemble with the ADP regularizer. The left part of each panel is the normalized non-maximal predictions.

equals to the squared volume spanned by the normalized non-maximal predictions, as illustrated in Fig. 1. Note that promoting the diversity still allows the maximal prediction of each network to be consistent with the true label, and thus will not affect ensemble accuracy. Besides, since the non-maximal predictions correspond to all potentially wrong labels returned for the adversarial examples, a high diversity or inconsistency on the non-maximal predictions of individual networks can lower down the transferability of adversarial examples among them, and further lead to better robustness of the ensemble.

Under our definition of ensemble diversity, we propose to promote the diversity using an *adaptive diversity promoting* (ADP) regularizer. The ADP regularizer consists of two terms—a logarithm of ensemble diversity (LED) term and an ensemble entropy term. The ensemble entropy term contains the Shannon entropy of ensemble predictions. In Section 3.3, we prove that the ensemble entropy term is necessary such that the LED term can properly encourage the diversity. The ADP regularizer encourages the non-maximal predictions of each member in the ensemble to be mutually orthogonal, while keeping the maximal prediction be consistent with the true label, as shown in Fig. 1 and theoretically proved in Section 3.3. In the training procedure, we use the ADP regularizer as the penalty term and train all the networks in the ensemble simultaneously and interactively on the same dataset. This is called the ADP training procedure.

In experiments, we test our method under several widely studied adversarial attacks introduced in Section 2.2 on MNIST (LeCun et al., 1998), CIFAR-10, and CIFAR-100 (Krizhevsky & Hinton, 2009). The results show that our method can significantly improve adversarial robustness while maintaining state-of-the-art accuracy on normal examples. Although when performing back-propagation (Rumelhart et al., 1986) in the ADP training procedure, we need to separately perform the operations $\det(G)$ and G^{-1} with the computational complexity of $\mathcal{O}(K^3)$, where K is the number of members in the ensemble, K usually grows much slower with the scale of problem (Russakovsky et al., 2015). This enables our method to scale to most of the classification tasks and avoid the computational obstacle caused by these matrix operations as encountered in previous work (Kulesza et al., 2012; Kwok & Adams, 2012; Mariet & Sra, 2016).

2. Preliminary Knowledge

In this section, we first define the notations, then briefly introduce the adversarial attacks that we test in experiments.

2.1. Notations

We denote a deep neural network (DNN) classifier as $F(x, \theta)$, where x is the input variable, and θ denotes all the trainable parameters. Let L be the number of classes, the output vector $F(x, \theta) \in \mathbb{R}^L$ represents a probability distribution (Hereafter we will omit θ without ambiguity). When training a classifier, one common objective function is the cross-entropy (CE) loss defined as

$$\mathcal{L}_{\text{CE}}(x, y) = -1_y^\top \log F(x) = -\log F(x)_y,$$

for an input-label pair (x, y). Here, 1_y is the one-hot encoding of y and the logarithm of a vector is defined as taking logarithm of each element. Given a probability vector F, the Shannon entropy is defined as $\mathcal{H}(F) = -\sum_i F_i \log(F_i)$.

2.2. The Adversarial Setting

Recent work shows that high-performance classifiers could still be fooled by adversarial examples (Kurakin et al., 2018b; Athalye et al., 2018). A generated adversarial example x^* is usually indistinguishable from its normal counterpart x by a human observer. Here we introduce some of the most commonly used attack methods as below:

Fast Gradient Sign Method (FGSM) (Goodfellow et al., 2015) crafts the adversarial example x^* as $x^* = x + \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(x, y))$, where $\mathcal{L}(x, y)$ is the adversarial loss.

Basic Iterative Method (BIM) (Kurakin et al., 2017a) iteratively crafts an adversarial example as $x_i^* = \text{clip}_{x,\epsilon}(x_{i-1}^* + \frac{\epsilon}{r} \cdot \text{sign}(\nabla_{x_{i-1}^*} \mathcal{L}(x_{i-1}^*, y)))$. Here $x_0^* = x$, r is the number of iteration steps and $\text{clip}_{x,\epsilon}(\cdot)$ is the clipping function.

Projected Gradient Descent (PGD) (Madry et al., 2018) has the same generation process as BIM, except for x_0^* is a randomly perturbed image in the neighborhood $\mathring{U}(x, \epsilon)$.

Momentum Iterative Method (MIM) (Dong et al., 2018) won the NeurIPS 2017 Adversarial Competition (Kurakin et al., 2018b), which is a variant of BIM. In each iteration, MIM updates the gradient g_i with momentum and crafts the

adversarial example as $x_i^* = \text{clip}_{x,\epsilon}(x_{i-1}^* + \frac{\epsilon}{r} \cdot \text{sign}(g_i))$

Jacobian-based Saliency Map Attack (JSMA) (Papernot et al., 2016b) perturbs one feature x_i by ϵ in each iteration step that maximizes the saliency map

$$S(x,y)[i] = \begin{cases} 0, \text{ if } \frac{\partial F(x)_y}{\partial x_i} < 0 \text{ or } \sum_{j \neq y} \frac{\partial F(x)_j}{\partial x_i} > 0, \\ (\frac{\partial F(x)_y}{\partial x_i}) \left| \sum_{j \neq y} \frac{\partial F(x)_j}{\partial x_i} \right| \text{,otherwise.} \end{cases}$$

The adversarial noise crafted by JSMA is often sparser, i.e., JSMA perturbs fewer pixels compared to other attacks.

Carlini & Wagner (C&W) (Carlini & Wagner, 2017b) define $x^*(\omega) = \frac{1}{2}(\tanh(\omega) + 1)$ in terms of ω, and minimize the objective function $\min_\omega \|x^*(\omega) - x\|_2^2 + c \cdot f(x^*(\omega))$, where c is a constant chosen by a modified binary search. Let $\mathbb{S}_{pre}(x)$ be the input vector of the softmax function in a classifier, then $f(\cdot)$ is the objective function defined as

$$f(x) = \max(\max\{\mathbb{S}_{pre}(x)_i : i \neq y\} - \mathbb{S}_{pre}(x)_i, -\kappa),$$

where κ controls the confidence on adversarial examples.

Elastic-net Attack (EAD) (Chen et al., 2018) is a variant of the C&W method, where the objective function becomes $\min_\omega \|x^*(\omega) - x\|_2^2 + \beta \|x^*(\omega) - x\|_1 + c \cdot f(x^*(\omega))$. EAD includes C&W as a special case when $\beta = 0$.

3. Methodology

In this section, we first introduce the training strategies for ensemble models. Then we propose the adaptive diversity promoting (ADP) training method, and provide theoretical analyses on the optimal solutions of the ADP method.

3.1. Training Strategies for Ensemble Models

In real-world problems, ensembling several individual classifiers is usually effective to improve generalization performance (Russakovsky et al., 2015). For clarity, we denote the k-th member in the ensemble as $F^k \in \mathbb{R}^L$, which represents the prediction of classifier k over L categories. Then a common construction of the ensemble prediction $\mathcal{F}$ is a simple average of all individual predictions as

$$\mathcal{F} = \frac{1}{K} \sum_{k \in [K]} F^k, \tag{1}$$

where $[K] := \{1, \cdots, K\}$. There are three major strategies for training ensembles, i.e., independent training, simultaneous training, and sequential training (Islam et al., 2003).

In independent training, each classifier F^k is trained separately, usually with the corresponding CE loss $\mathcal{L}_{\text{CE}}^k(x, y) := -\log F^k(x)_y$. Because of its simplicity, most of the deep learning systems apply this strategy (Russakovsky et al., 2015; Liao et al., 2018). However, one major disadvantage of independent training is that it ignores the interaction among the members in the ensemble. This ignorance may lead to similar predictions or shared feature representations among these members (Dauphin et al., 2014; Li et al., 2016), which would not be useful to improve the performance of the ensemble (Krogh & Vedelsby, 1995), and result in less improvement on robustness as shown in our experiments.

In simultaneous training, all the classifiers are trained on the same mini-batch of data in each training iteration. A naive objective function for simultaneous training is the ensemble cross-entropy (ECE) loss defined as

$$\mathcal{L}_{\text{ECE}} = \sum_{k \in [K]} \mathcal{L}_{\text{CE}}^k, \tag{2}$$

which is a simple sum of all individual cross-entropy losses. Since the parameters for different networks are typically independent, simultaneously training all K networks with the ECE loss is equivalent to independently training each classifier with the corresponding CE loss. The learning rate for each network could be different. If some networks converge faster, then we can freeze the parameters of those converged networks and continue training others. Simultaneous training scheme allows us to incorporate the interaction among different members in the ensemble. Indeed, for large-scale tasks, simultaneous training may require more GPU memory or workers compared to independent training. However, it does not require extra computation. When the computing resource is limited, sequential training can be applied instead, which trains one network after another in each iteration.

In our method, we apply the simultaneous training scheme. Namely, we combine a novel regularizer (introduced in Eq. (5)) with the ECE loss term as the objective function for training. By doing this, the members in the ensemble can separately keep high accuracy since the ECE loss term, while interacting with each other via our regularizer.

3.2. Adaptive Diversity Promoting Training

Previous work has shown that for a single network, promoting the diversity among learned features of different classes can improve adversarial robustness (Pang et al., 2018a;b). In this paper, we promote diversity from another level, i.e., the diversity among learned predictions of different members in an ensemble, to improve robustness. This kind of diversity is called *ensemble diversity*. The previous analysis states that there is no strict definition of what is intuitively perceived as ensemble diversity (Kuncheva & Whitaker, 2003) and most of the previous work defines the ensemble diversity concerning the prediction errors of individual classifiers in the ensemble (Liu & Yao, 1999a;b; Dietterich, 2000; Islam et al., 2003). This definition partly bases on the assumption that each member in the ensemble is a weak classifier (Friedman et al., 2001). However, this assumption hardly holds for the ensemble of DNNs nowadays, thus encouraging the prediction errors of DNNs to be diverse is inappropriate and will largely sacrifice the accuracy of them.